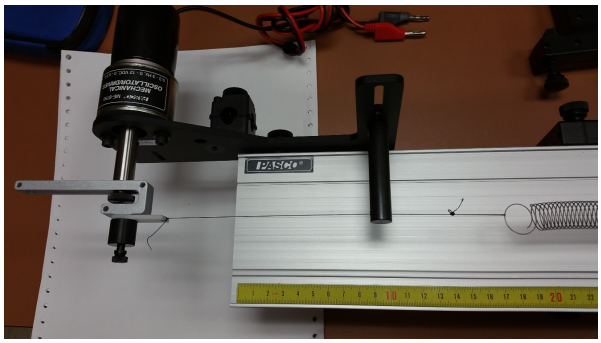
# Lab 5: Forced vibration

**Purpose:** Motors and pumps can amplify vibration near resonance. Measure the cart response to a brief tap and to a motor-driven spring end; compare with the model using your measured parameters.

**Before your section:** Complete Pre-Lab 5. **Combined Labs 4 + 5 report:** Monday, November 23, 11:59 PM. Follow Canvas for submission.




Course apparatus: the motor moves one spring end; the Vernier cart records its own position.

## A. Measure and connect

**1.** Record the cart mass, installed hex rods and all damper parts that move with the cart. Add them to obtain total moving mass **m [kg]**. Record the spring IDs and their Lab 4 stiffnesses: $\mathbf{k_1}$ on the fixed side and $\mathbf{k_2}$ **[N/m]** on the motor side. Use the Lab 4 spring package if a spring has not been calibrated.

**2.** Level the track and attach the two springs securely. Keep both taut throughout the motion, near their course working length of about 15 cm. Leave clearance from the end stops. Keep the motor OFF while changing connections.

**3.** Turn on the cart. Open **Vernier Graphical Analysis** > **New Experiment > Sensor Data Collection**. Select the cart by its serial number, then Connect / Done. Display **Position (m) versus Time (s)**.

**4.** With the cart resting at equilibrium, select **Position > Zero**. Set collection to **25 samples/s**, or the GTA-approved lower rate. Check that the motion is resolved; ask the GTA if it is not. Zero again after repositioning the cart.

## B. Record three tap responses

**5.** Remove magnetic dampers and keep both spring ends stationary. Start **Collect** with the cart resting at equilibrium. Give the cart body **one brief, gentle horizontal tap with your fingers**, then remove your hand immediately. Record several clean oscillations until the motion is small.

**6.** Repeat from rest with light, medium and slightly stronger gentle taps in the same direction. Keep mass and springs unchanged. Stop and export each run separately: **document/file menu > Export > CSV > Save CSV**. Name them Tap_1.csv, Tap_2.csv and Tap_3.csv.

**Tap versus Lab 4 release:** A tap mainly gives initial velocity; pull-and-release starts from a displaced position without pushing. Tap strength is qualitative because force is not recorded. Do not strike an end stop or push continuously.

# Calibrate the input; collect the sweep

## C. Measure damping and motor travel

**7.** Add the assigned **1-2 magnetic dampers** and remeasure total moving mass. With the motor OFF and both outer spring ends held stationary, record a small pull-and-release decay. Analyze it with MATLAB **mode 1** (next page) to obtain **c [N s/m]**. Use this damping only with the same mass, springs and damper configuration.

**8.** With the GTA, measure the full travel of the motor-driven spring endpoint. Use a ruler while the drive is stopped at its two extreme positions, following the GTA's setup procedure. Target about **2-3 cm peak-to-peak**; record the actual travel. Never adjust a moving linkage. If the station uses direct-hook cart calibration, have the GTA make that temporary connection.

**9.** Convert measured endpoint travel to meters. The input peak amplitude **Y** is half the peak-to-peak travel: 2.4 cm peak-to-peak means 0.024 m entered in MATLAB and Y = 0.012 m. This is endpoint motion, not cart motion. Recheck it if the pin setting or travel changes.

## D. Record the harmonic response

**10.** Reconnect the cart between the fixed-side and driven-side springs. Keep the dampers and mass from step 7. With the GTA, increase the motor supply gradually within the course setup range **2-11 V**. Measure actual motor speed using the digital tachometer; voltage is not frequency.

**11.** For one drive cycle per revolution, drive frequency **f [Hz] = rpm/60**. Natural frequency from the measured mass and springs is $\mathbf{f_n = \sqrt{[(k_1 + k_2)/m]}/(2\pi)}$. Plan **3-5 drive frequencies below and 3-5 above** the resonance region, with GTA approval. Do not dwell at resonance.

**12.** At each speed, wait until the amplitude is approximately stable (often 1-2 minutes; judge from the graph). Save at least **five settled cycles, preferably ten**. To plan duration, ten cycles take 10/f seconds. Keep a startup/transient record for at least one setting so you can show the transition.

**13.** Record rpm and input peak-to-peak travel. Stop collection, export one CSV per speed, and note the start/end times of the settled interval from the Vernier graph. If amplitude grows excessively, stop the motor. If a spring goes slack, speed drifts or the cart approaches an end stop, correct the setup with the GTA and repeat.

**Run log:** filename, total moving mass, both spring IDs/stiffnesses, magnet count, matching c, measured rpm, input peak-to-peak travel [m], and settled start/end times [s]. Use MIE402_Lab5_Run_Log.csv. Open one exported file to confirm that time and position were saved.

**Keep the comparison controlled:** Do not change mass, springs or magnets during one sweep. If you change them, record a matching free decay and start a separately labeled sweep.

# Analyze in MATLAB; prepare the report

## E. Run the supplied analysis

**14.** Download **MATLAB files > Complete bundle** from the Lab 5 page. Unzip it and keep the three analysis .m files together. Open **Run_Lab5_Forced_Response.m** and click Run; do not run helper files individually.

**15.** For taps or the damping-calibration decay, choose **1 = impulse/free decay**. Enter your total moving mass [kg] and the two individual stiffnesses [N/m]. Select the run CSV/Excel files. For Excel, choose the correct worksheet. Enter time/position column numbers, e.g. [1 2], or [] for clear headers; an index column may shift them to [2 3].

**16.** Enter the clean start/end times from Vernier, **after hand contact ends**, and choose **1 = meters**. Check that the fitted curve follows the measured points. For the damped calibration, read c_est_Ns_m from Lab5_Impulse_Summary.csv and use it for the matching sweep.

**17.** For motor runs, choose **2 = harmonic sweep**. Enter matching mass, stiffnesses and c. Select the files. For each file, enter the settled start/end times, meters, measured rpm and input **peak-to-peak travel in meters**. The code halves this travel to obtain Y; do not halve it before entering.

**18.** Open the exported **Lab5_Results** folder beside your data. Review each measured trace, sine fit and analytical-amplitude curve, then Lab5_Frequency_Response.png. If a fit misses the points, check columns, units, rpm and the chosen interval before accepting the result.

## F. What the comparison means

$$m\ddot{x} + c\dot{x} + (k_1 + k_2)x = k_2 y(t)$$

$$\frac{X}{Y} = \frac{k_2}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}$$

**x(t)** is cart displacement and **y(t) [m]** is spring-end displacement. **X [m]** is the fitted cart peak amplitude. Compare measured **X/Y** with theory using your m, stiffnesses and c. The plot uses 20 $\log_{10}$(X/Y) versus angular drive frequency **ω = 2πf [rad/s]**. Use fitted amplitudes, not FFT peak heights. The analytical time curve is phase-aligned; input/output phase was not measured.

## G. Include in the combined Labs 4 + 5 report

**Tap tests:** Three time histories in one figure and three spectra in another; label measured peak frequencies. Discuss whether stronger taps change amplitude or frequency.

**Motor tests:** One startup-to-settled record with the fitted interval marked; one representative settled data/fit/theory comparison; the measured frequency-response points overlaid on theory; and a compact run table with f, X, Y and X/Y.

**Connect to Lab 4:** Distinguish free-decay frequency from imposed drive frequency. Explain departures from theory using your records. Include only equations used in the analysis.

## Before leaving

Show the GTA one tap analysis, one settled harmonic fit and a frequency-response plot with at least five measured points. This is a bench check, not a replacement for the full planned sweep. Share raw files, calibration records, run log and MATLAB exports with every group member.